



Review Article

The role of AI in emergency department triage: An integrative systematic review

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Abstract

Background

Overcrowding in emergency departments (EDs) leads to delayed treatments, poor patient outcomes, and increased staff workloads. Artificial intelligence (AI) and machine learning (ML) have emerged as promising tools to optimize triage. Objective: This systematic review evaluates AI/ML-driven triage and risk stratification models in EDs, focusing on predictive performance, key predictors, clinical and operational outcomes, and implementation challenges.

Methods

Following PRISMA 2020 guidelines, we systematically searched PubMed, CINAHL, Scopus, Web of Science, and IEEE Xplore for studies on AI/ML-driven ED triage published through January 2025. Two independent reviewers screened studies, extracted data, and assessed quality using PROBAST, with findings synthesized thematically.

Results

Twenty-six studies met inclusion criteria. ML-based triage models consistently outperformed traditional tools, often achieving AUCs > 0.80 for high acuity outcomes (e.g., hospital admission, ICU transfer). Key predictors included vital signs, age, arrival mode, and disease-specific markers. Incorporating free-text data via natural language processing enhances accuracy and sensitivity. Advanced ML techniques, such as gradient boosting and random forests, generally surpassed simpler models across diverse populations. Reported benefits included reduced ED overcrowding, improved resource allocation, fewer mis-triaged patients, and potential patient outcome improvements.

Conclusion

AI/ML-based triage models hold substantial promise in improving ED efficiency and patient outcomes. Prospective, multi-center trials with transparent reporting and seamless electronic health record integration are essential to confirm these benefits.

Implications for Clinical Practice

Integrating AI and ML into ED triage can enhance assessment accuracy and resource allocation. Early identification of high-risk patients supports better clinical decision-making, including critical care and ICU nurses, by streamlining patient transitions and reducing overcrowding. Explainable AI models foster trust and enable informed decisions under pressure. To realize these benefits, healthcare organizations must invest in robust infrastructure, provide comprehensive training for all clinical staff, and implement ethical, standardized practices that support interdisciplinary collaboration between ED and ICU teams.

Introduction

Emergency departments (EDs) play a pivotal role in the global healthcare system by providing immediate medical attention to individuals with acute and often life-threatening conditions [1]. Despite their critical importance, EDs worldwide continue to grapple with persistent overcrowding, lengthy patient waiting times, and resource strain—factors that collectively compromise the quality and timeliness of emergency care [1,2]. This chronic state of overcrowding is particularly troubling because timely triage and intervention can significantly impact patient outcomes, such as mortality,

length of hospital stay, and overall morbidity [[1], [2], [3]]. Consequently, effective and efficient triage mechanisms have become central to efforts aimed at optimizing ED operations.

Triage is the systematic process of rapidly categorizing patients based on the urgency of their condition to prioritize immediate care. Globally, emergency departments use rigorously validated systems—such as the U.S. five-level Emergency Severity Index (ESI), Europe's Manchester Triage System (MTS), as well as the Canadian Triage and Acuity Scale (CTAS) and the Australasian Triage Scale (ATS)—to swiftly stratify patient acuity. Despite these regional variations, all these systems share the goal of early identification of high-risk patients to ensure timely, effective intervention [[4], [5], [6], [7]]. These systems classify patients into discrete categories corresponding to acuity, guiding resource allocation and the order in which patients are seen. However, conventional triage scales do not fully account for broader operational factors, such as fluctuating ED bed capacity or surge demands [8].

While the traditional systems provide a structured approach to patient assessment, their static nature and reliance on predefined categories often limit their ability to dynamically adapt to fluctuations in patient volume and the evolving complexity of clinical presentations. In response to these challenges, recent advancements in artificial intelligence (AI) and machine learning (ML) have emerged as promising avenues to enhance triage processes [9].

In response to these challenges, recent advancements in AI and ML have emerged as promising avenues to enhance triage processes. By harnessing vast amounts of structured data (e.g., vitals, labs) and unstructured data (e.g., chief complaints, nursing notes), AI-driven models offer the potential to improve prediction accuracy and operational efficiency, thereby addressing some of the inherent limitations of traditional approaches [[10], [11], [12], [13]].

Despite these promising developments, the evidence base remains fragmented due to heterogeneity in study quality, settings, and patient populations, underscoring the urgent need for a systematic synthesis of quantitative performance metrics and qualitative implementation insights. Decisions made during ED triage can directly influence ICU admission rates and patient outcomes, demonstrating a critical link between emergency care and the intensive care environment [14]. Timely identification of critically ill patients at triage enables urgent intervention and appropriate ICU admission, which is central to prevent deterioration and reducing mortality[15,16]. Moreover, these triage decisions have downstream effects on critical care workflows, as appropriate ED triage triggers the mobilization of ICU resources and personnel, whereas mis-triage or capacity strain can compromise rapid response efforts and intensive care preparedness[[17], [18], [19]]. Our review provides critical care nurses with valuable insights into how AI-driven triage innovations can facilitate early ICU admissions, optimize resource allocation, and ultimately enhance outcomes for severely ill patients.

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Section snippets

Aim

To critically evaluate and synthesize the current evidence on the validation and clinical impact of AI and ML-based models for triage and risk stratification in emergency departments—encompassing both general and disease-specific applications—with an emphasis on performance metrics, workflow feasibility, and patient outcomes. ...

Objectives

1. To compare the accuracy and discrimination of AI/ML-based triage and risk stratification models with conventional triage methods (e.g., ESI, MTS) or scoring tools. ...
2. To identify the most influential predictors (structured and unstructured data) used in AI/ML triage models across various patient populations. ...
3. To evaluate the reported effects of AI/ML-driven triage tools on ED workflows (e.g., impact on decision-making efficiency, wait times, and resource allocation) and on key patient outcomes (e.g., ...

...

Methods

This systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [20], ensuring transparency and replicability. Because literature features both quantitative and mixed-methods studies, we used a thematic synthesis approach with thematic grouping to capture performance metrics and broader contextual insights.

We employed the SPIDER (Sample, Phenomenon of Interest, Design, Evaluation, Research type) framework [21], to guide ...

Information sources and search strategy

A comprehensive literature search was executed across multiple electronic databases to ensure exhaustive retrieval of relevant studies from inception up to January 2025. The databases searched are PubMed, CINAHL, Scopus, Web of Science, and IEEE Xplore. Additionally, the reference lists of included studies and pertinent reviews were hand-searched to identify any additional relevant studies. The search strategy employed a combination of keywords and Medical Subject Headings (MeSH) terms related ...

Selection process

The records identified through comprehensive database searches were imported into Rayyan, a specialized systematic review screening platform [22]. To ensure methodological rigor and minimize bias, two independent reviewers conducted the screening process. Any discrepancies were addressed through consensus discussions. The protocol for resolving discrepancies involved detailed discussions to reach consensus, ensuring that inclusion decisions were unbiased and based on predefined criteria. The ...

Data extraction and analysis

Data extraction was performed using a standardized form. This form was designed to capture comprehensive details from each study and included the following components: (1) Reference Details – including authors, publication year, and journal; (2) Research Design – specifying whether the study was retrospective, prospective, cross-sectional, etc.; (3) Study Aim and Objectives; (4) Population and Setting – detailing the sample size, geographical context, and specific ED characteristics; (5) AI/ML ...

Risk of bias and quality of included studies

To systematically evaluate the methodological rigor and risk of bias of the included studies, the Prediction Model Risk of Bias Assessment Tool (PROBAST) was employed [24]. PROBAST is specifically designed for assessing the quality and applicability of studies developing, validating, or updating prediction models. It evaluates four critical domains: Participants, Predictors, Outcomes, and Analysis, focusing on potential sources of bias and the generalizability of findings.

Each included study ...

Study selection and characteristics

The initial search yielded 334 records. After removing duplicates and screening titles/abstracts, 56 full-text articles were assessed for eligibility, and 26 met the final inclusion criteria (Fig. 1). The

included studies span 2018 to late 2024, drawn from diverse geographical regions (North America, Asia, Europe, Africa, Middle East, South America). Designs were predominantly retrospective, with a few prospective or partially prospective validations. (See Appendix 1 for a summary of each ...

Evaluating the risk of bias and quality of included studies

The systematic evaluation of included studies using PROBAST reveals a predominantly moderate risk of bias across research focused on ED triaging using ML and Large Language models (LLMs). This assessment spans four key domains: Participants, Predictors, Outcomes, and Analysis. ...

Predictors

The predictors domain consistently demonstrated a low risk of bias across studies. Predictors included standardized clinical parameters, such as vital signs, demographics, and chief complaints, which are routinely collected during ED triage. Furthermore, advanced techniques like Natural Language Processing (NLP) were employed in several studies to analyze free-text chief complaints, ensuring relevance and reliability. The consistent use of these robust and readily available predictors across ...

Outcomes

The Outcomes domain showed a moderate risk of bias in most studies. While clinically meaningful outcomes—such as hospital admissions, ICU admissions, and mortality—were used, their definitions often depended on institution-specific practices and protocols. For example, disparate criteria for hospital admissions or critical care allocation likely introduced variability, limiting the consistency and comparability of outcomes. Furthermore, the retrospective nature of most studies compounded the ...

Analysis

In the Analysis domain, a moderate risk of bias was identified across studies. While most studies employed robust methodologies, including cross-validation, multiple algorithm comparisons, and comprehensive performance metrics (e.g., AUC, sensitivity, specificity), the lack of external validation was a pervasive limitation. For example, Lin et al. [27] conducted external validation to enhance generalizability, whereas Leonard et al. [28] and Ivanov et al. [29] relied solely on internal ...

Overall risk of bias and applicability

Synthesizing the results across domains, the overall risk of bias for the included studies was predominantly moderate. Single-center designs, selection biases, and the lack of external validation were the main contributors. These limitations highlight the need for future studies to prioritize multi-center collaborations, standardized outcome definitions, and external validation to enhance the generalizability and clinical utility of AI/ML-based triage models. (See Appendix 2). ...

Thematic synthesis

The comprehensive analysis of the included studies revealed four primary themes that encapsulate the integration and impact of AI and ML technologies in emergency department ED triage and risk stratification: Predictive Performance, Key Model Drivers, Clinical and Operational Impact, and Implementation Feasibility. These themes collectively illustrate the multifaceted benefits and challenges associated with deploying AI/ML-based triage tools in diverse ED settings. Table 4 provides a concise ...

Accuracy and discrimination across diverse models

Out of the 26 studies, 14 reported AUC values above 0.80 [26,28,[30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]] for outcomes such as hospital admission or critical care need, often exceeding the performance of tools like ESI or Quick Sequential Organ Failure Assessment (qSOFA). However, these AUCs reflect retrospective data and may not translate directly to prospective, real-time accuracy.

ChatGPT-based triage [25] recorded a Cohen's Kappa of 0.828 for high-acuity ...

Theme 2: Key model drivers

Vital signs (heart rate, respiratory rate, blood pressure, temperature, oxygen saturation) were near-universal predictors. Demographics (especially age) and arrival mode (ambulance vs. walk-in) often emerged as robust features. In disease-specific contexts (sepsis, chest pain, TBI), lab markers (e.g., CRP, creatinine) or clinical scales (e.g., Glasgow Coma Scale) offered added predictive power. Textual features extracted via NLP also improved classification. For instance, keywords like “stroke” ...

Effects on ED overcrowding and resource allocation

ML models aimed to reduce ED overcrowding by proactively identifying admission risk or critical illness, thereby expediting bed assignment and streamlining care pathways. Out of the 26 included studies, 9 studies [26,[28], [29], [30],32,35,42,44,45] explicitly reported that early admission prediction or improved triage accuracy optimized workflow by enabling targeted resource allocation, reducing ED boarding, or decreasing overcrowding. ...

Decision-Making efficiency and staff workload

AI-based triage solutions, including ChatGPT [25] and ...

Integration with EHRs and Real-Time systems

Although most studies were retrospective, a few [25,40,46] demonstrated partial prospective components or near-real-time validation. EHR integration remains a critical facilitator; however, variable data quality and missing values often limited immediate adoption [26,29]. Limited infrastructure and data quality remain major barriers. ...

Interpretability and explainable AI

Several teams employed SHapley Additive exPlanations (SHAP) [40,45] or rule-based decision trees to enhance interpretability [34,43]. Such transparency fosters ...

Discussion

This discussion synthesizes findings from 26 studies organized into four themes: predictive performance, key model drivers, clinical and operational impact, and implementation feasibility. Our review found that AI/ML-based triage models frequently report high predictive accuracy (often with AUCs above 0.80), suggesting improved identification of high-acuity patients and more efficient resource allocation compared to traditional systems. However, the predominance of retrospective, single-site ...

Strengths

This systematic review employed a broad and robust approach by searching multiple major databases—PubMed, CINAHL, Scopus, Web of Science, and IEEE Xplore—and by hand-searching reference lists, thereby reducing the likelihood of missing important studies. The use of the SPIDER framework facilitated an inclusive yet structured process, accommodating both quantitative and qualitative research designs relevant to AI/ML-based triage. In addition, risk of bias was evaluated using the PROBAST tool, ...

Limitations

Despite the breadth of the search strategy, most of the included research comprised single-center, retrospective studies with moderate overall risk of bias, thus limiting the generalizability of the findings. In our review, although studies originated from multiple regions, 9 of 26 studies (35%)

were conducted in the United States and 5 of 26 (19%) in China, indicating that more than half of the evidence is derived from these two countries. This geographic concentration may limit the ...

Recommendations and implications

While the promise of AI/ML in ED triage is clear, we must temper optimism with the lessons learned from past health IT implementations, especially with EHRs. Historically, even highly accurate models have struggled when integrated into the complex, human-driven clinical environment. Thus, innovative algorithms alone will not transform triage; their success hinges on careful, systematic integration into clinical workflows. Importantly, these triage decisions directly affect ICU admissions and ...

EHR integration – Necessary but not Sufficient

Integrating AI triage models into existing EHR systems is a vital step toward real-time clinical decision support. However, this integration alone will not resolve all challenges. Many ED datasets suffer from variable data quality, completeness, and standardization. For example, several studies noted that missing or unstructured data hindered model performance. Therefore, future efforts must emphasize robust data exchange protocols and uniform data standards—such as adopting common ontologies ...

Addressing publication bias

A notable gap in the literature is the absence of reporting on positive predictive value (PPV) or related metrics that inform how many patients flagged as high-risk actually benefit from urgent intervention. None of the 26 studies provided these measures, which are essential for assessing the real-world clinical utility of these tools. The selective publication of studies with positive findings likely overestimates the efficacy of AI models. To mitigate this bias, future reviews should extend ...

Real-World implementation challenges

Beyond integration and publication bias, several practical challenges must be addressed for AI-based triage systems to succeed:

- **Generalizability:** Many algorithms were developed and validated in single-hospital settings. Future research must include prospective, multi-center trials that test these models in various ED environments—academic and community hospitals across different regions and patient demographics. ...
- **Clinician Trust and Acceptance:** Critical care nurses and other frontline clinicians ...

...

Strategic Directions for future research

Based on our findings, we propose the following key strategies:

1. **Prospective, Multi-Center Validation:** Future studies must conduct rigorous prospective trials across multiple ED sites, adhering to guidelines such as CONSORT-AI and SPIRIT-AI, to validate model performance in varied real-world conditions. ...
2. **Comprehensive Performance Reporting:** Researchers should expand reporting beyond AUC, sensitivity, and specificity to include clinically actionable metrics such as PPV, false positive rates, and ...

...

Implications for critical care nursing

As ED triage directly influences ICU admissions, the integration of AI models has significant implications for critical care nursing. Improved triage accuracy can streamline the transition of critically ill patients to the ICU, reduce delays in intervention, and ultimately enhance patient outcomes. Engaging critical care nurses in the design and implementation process is essential to ensure that AI tools support their clinical workflows, reduce workload, and foster interdisciplinary

...

Conclusion

AI and ML-based triage models show promise in retrospective comparisons with conventional triage, often reporting superior predictive metrics for critical outcomes. By incorporating structured and unstructured data, these tools could potentially reduce overcrowding, optimize resource allocation, and improve patient outcomes. However, the current evidence base lacks robust prospective validations and multi-center trials, making it premature to assert definitive impacts on mortality or ED ...

CRedit authorship contribution statement

Rabie Adel El Arab: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Omayma Abdulaziz Al Moosa:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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